

The 4Ps: How Might We Help Cities Better Learn What Their Residents Need?

This document contains learnings from

- 11 stakeholder interviews (DSS, NYC Homeless Services, Career & Training Services, LA homeless outreach, Roosevelt Island/RIOC, the Manhattan Borough President's Office, an engaged 311 user, a makerspace nonprofit, and two advisors), and
- Outside research from government sources and academic literature

Players

Who is in the industry? Who holds power, and why?

City-level intake & service agencies

- **NYC 311**: Centralized, non-emergency clearinghouse for city government. Originally proposed by Mayor Bloomberg in his 2002 State of the City address, it consolidated more than 40 separate agency call centers and 12 pages of phone-book listings into a single number, launching March 9, 2003. It is run out of the Mayor's Office of Operations with technology support from the city's IT agency, and Joseph Morrisroe (our upcoming interview) has led the 311 operation since 2006. (*Interview: 311NYC session, Team notes 9/10; NYC.gov 311 Best Practice report; podcast interview with Morrisroe*)
- **NYC Department of Social Services (DSS)**, comprised of the Human Resources Administration (HRA) and Department of Homeless Services (DHS) under one consolidated management structure — the largest local social-services agency in the country, serving roughly 3 million New Yorkers a day across Cash Assistance, SNAP, Medicaid, rental assistance/homelessness prevention, and related programs. (*Interview: Pinto, DSS food stamp department, 9/11; NYC DSS/HRA Preliminary Budget Testimony*)
- **NYC Homeless Services Department / Office of the Ombudsman** — the escalation point when a resident's issue isn't resolved elsewhere; staffed by people like Mr. Vargas who personally connect residents to IT support (e.g., "Luis") when systems don't talk to each other. (*Interview: Mr. Vargas, 9/11*)
- **Career and Training Services** — a satisfaction-oriented service point; front-line staff (security guard interviewed) report people leave happy once they engage. (*Interview, 9/11*)

Elected & appointed local government

- **Manhattan Borough President's Office (MBPO)** — an independent city agency of 50+ staff that acts as convener and advocate rather than legislator; it is *advisory* but influential, holding mandated recurring meetings with city agencies. The Borough President appoints ~300 unpaid Community Board members per year across Manhattan's 12 boards (25 nominated by City Council members, 25 by the Borough President directly), and chairs the Borough Board. (*Interview: Andrew Chang & Marielle Ali, 9/4; NYC Charter §82(17)(a); MBPO Community Board demographic reports; Wikipedia/NYC.gov Community Board structure*)
- **Community Boards (59 citywide, 12 in Manhattan)** — locally appointed advisory bodies mandated by the NYC Charter (§2800) that review land use/zoning, comment on the city budget, and address municipal service delivery; a public but under-recognized layer of governance ("not that many people know about them," per interview). (*Interview 9/4; NYC.gov Community Board overviews*)
- **City Council members** — nominate half of Community Board appointees and receive resident complaints directly.

- **Roosevelt Island Operating Corporation (RIOC)** — governs the state-leased land Roosevelt Island sits on (lease extended to 2078); Jane (Cornell Tech's community liaison since 2014) coordinates Cornell Tech's civic presence with RIOC, the Senior Center, the Hospital, Four Freedoms Park, and PSI School. (*Interview: Jane, 9/10*)

Non-profit & community-based organizations

- **St. Joseph's Center (LA)** — government subcontractor doing street outreach/case management; Erik, our interviewee, self-describes as a "humanitarian salesman" convincing unhoused residents to use a system they don't trust. (*Interview: Erik, 9/10*)
- **LAHSA (Los Angeles Homeless Services Authority)** — the lead homelessness agency for LA City/County, subcontracting to smaller service providers; central to the "trust issues" theme in our LA interview. In June 2026, HUD suspended LAHSA's federal Continuum of Care funding after an Office of Inspector General investigation found LAHSA could not verify occupancy of transitional-housing motel rooms, misused funds meant for other contracts, and lacked documentation for nearly 2,300 housing sites it was responsible for — on top of the \$300M in county funding pulled from LAHSA in 2025 after conflict-of-interest findings involving its former CEO. HUD's Office of Inspector General opened an investigation and found evidence of repeated false statements, poor financial controls, and inadequate safeguards against conflicts of interest, including inability to verify whether funds were paying for empty hotel rooms, misuse of funds meant for other government contracts, and missing documentation for nearly 2,300 housing sites. This is a live case study in what happens to resident trust when the *data layer* (occupancy tracking, spend reporting) breaks down. (*Interview 9/10; HUD press release, June 2026*)
- **Community Service / civic clubs at Cornell Tech** — driven largely by PhD students, coordinate Halloween events, Roosevelt Island Disabled Association food distribution, "1-to-1 Tech Days" for older adults, and food pantry events. (*Interview: Jane, 9/10*)
- **Makerspace nonprofits** (e.g., SF Makerspace, referenced by Curren Mandon) — run certification programs, membership tiers, and business accelerators funded through recurring grant cycles. (*Interview: Curren Mandon, 9/9*)

Residents / constituents

- Engaged citizens who already know how to navigate the system (e.g., Robbie Dornbush, who used 311 successfully for a flooding hydrant and separately contacted a member of Congress about polling-place communications). (*Interview, 9/10*)
- Digitally hesitant residents who "don't want to learn the technology" and want someone to do it for them (per Mr. K). (*Interview, 9/11*)
- Non-English speakers who need interpreters DSS doesn't have enough of. (*Interview: Pinto, 9/11*)
- Unhoused residents in shelters, who — per Mr. Vargas — sometimes prefer the shelter system because it offers human connection they lack elsewhere, and per Erik, sometimes decline help outright.

Academic/data-research players

- NYU's Marron Institute of Urban Management / Civic Analytics Program and NYU's Center for Urban Science and Progress, which studies exactly the bias questions our project cares about (see Pain Points/Processes below). Constantine Kontokosta directs the Civic Analytics program at the NYU Marron Institute of Urban Management and holds cross-appointments at the Center for Urban Science and Progress and NYU Tandon's Department of Civil and Urban Engineering.
- Northeastern University's Boston Area Research Initiative (Daniel T. O'Brien, author of *The Urban Commons*).
- MIT political science (Ariel White & Kris-Stella Trump), authors of a widely cited 311-data study.

Upcoming interviews to fill Players gaps:

- **Gayan Seneviratna (MTA, Data & Analytics division)** — will help us understand the *transit* side of the "players" map, an agency we currently have no first-hand insight into.
- **Joseph Morrisroe (Deputy, NYC Office of Technology & Innovation, 311)** — will let us go directly to the top of the 311 chain of command rather than relying on secondary sources for how 311 leadership thinks about data and routing.

Processes

How does the work happen? Where does it wait, break, or get repeated?

Resident-to-government process

Described during meeting

1. A resident raises a concern through whatever channel is available to them — in person, phone call, email, typed or handwritten letter, or social media tag. Channel choice is "very person dependent."
2. If unresolved, the resident escalates to local elected officials, community boards, or a city office — often via manual research (Google search, social media).
3. MBPO staff triage: *Who are the stakeholders? Who has power to resolve this? What laws apply? Where does enforcement sit?* Legal/court matters are explicitly out of scope for their office.
4. Staff propagate the issue to the relevant stakeholder(s) and research history — has this happened before? Is it a one-off (e.g., a pothole) or a recurring/long-term issue (e.g., a chronic noise complaint)?
5. High-level issues may go all the way to the Mayor's Office.
6. Residents are told to log recurring issues through 311 so the frequency shows up in city data — the *only* real feedback loop city agencies systematically watch. (*Interview 9/4*)

311 Additional Information

- **311 specifically** works as a **call/intake center that federates requests out to owning agencies** rather than resolving them directly — a Parks tree issue is routed to Parks, a DEP issue to Environmental Protection, etc., which requires tight backend coordination among agencies. 311 operates primarily as a call center for public requests and federates the request to the appropriate agency — for example, tree-related requests are directed to Parks — which requires the agencies to work closely together so the backend technology stays seamless. The system now fields more than 42 million individual contacts per year, more than any other municipal non-emergency line in the country. Each call is walked through a structured script and, historically, agents cross-referenced a knowledge base of roughly 7,000 topics to route the caller correctly. The resulting complaint-level data is aggregated by the city for identifying citywide patterns and, since 2008, was formalized into public Citywide Performance Reporting dashboards.
- **Academic framing of the 311 "process" as a measurement instrument:** two studies are directly relevant to how we should interpret any process built on resident self-reporting. 311 call data is empirically and theoretically different from other citizen-participation measures — rates of 311 calls are negatively related to lower-cost participation like voter turnout and census return, but positively related to the higher-cost

activity of campaign donation, so researchers caution against treating 311 volume as a generic measure of political engagement. Separately, Boston-based research on 311 "custodianship" finds that 311 reports do not fully describe community needs, since residents in some neighborhoods are more proactive about reporting shared problems than others, and that people who report to 311 are no more likely to vote, volunteer, or otherwise engage civically than anyone else. This directly supports the Advisor and Coach's "watch, don't ask" MVP framing.

Homeless-services process (LA case study, transferable for NYC DHS):

1. Requests enter through LA's **Homeless Outreach Portal**, get assigned to outreach teams of case managers — described as "nothing proactive... except emergency."
2. Requests originate from residents, businesses, police, council-district offices, or advocates.
3. Case managers triage by neighborhood ("Westwood day"), then perform case management: obtaining documents for housing/mental health, making referrals, waiting for a housing "match," and driving clients to appointments.
4. The system is **entirely referral-based** — outreach workers individually refer each case rather than working from a shared, proactive view of need. *(Interview: Erik, 9/10)*
5. Data systems **HMIS Clarity and Champ** collect houseless-persons data; Clarity has recently added a feature letting clients view their own profile. *(Interview 9/10 — worth probing further with a data/analytics contact, potentially relevant to our MTA conversation as a model of a client-facing data layer.)*

NYC Department of Human Services

- **NYC DSS/food stamps intake process:** described by Pinto as functionally complete (most residents who show up get what they need) but bottlenecked by **staffing, not technology**. Mentioned a potential shortage of interpreters as an afterthought *(Interview 9/11)*
- **NYC Homeless Services Ombudsman process:** breaks down less because policy is missing and more because **systems don't talk to each other** ("issues with the systems updating, not compatible"), Leadership, per Vargas, doesn't "start from the ground up," so official awareness of real conditions lags what front-line staff see, and "wins" go unreported upward. *(Interview 9/11)*

Civic Engagement Processes

- **Cornell Tech ↔ Roosevelt Island community engagement process:** driven almost entirely by one person (Jane) with no staff — she maintains a list of ~900 contacts in Google, blasts relevant opportunities by email, and personally attends recurring meetings (hospital task force, Senior Center). There is **no formal tracking process** for relationships, no handover documentation system, and much of Cornell Tech's actual

community engagement happens through informal faculty relationships Jane isn't even aware of. (*Interview 9/10*)

Profits

Where does the money come from? Where does it go? Who carries the risk?

- **NYC DSS/HRA** is the most fiscally transparent player we've encountered. DSS is the largest local government social-services agency in the country, combining HRA and DHS to serve roughly 3 million New Yorkers a day, with a budget in which Medicaid consumes 64% of the city-funded portion, 15% goes to Public Assistance, 11% to rental assistance and homelessness prevention, 3% to administration, 2% to legal services, and 1% each to HIV/AIDS services, employment services, food assistance, and domestic-violence services. The agency has historically operated with more than 14,000 staff and an operating budget in the range of \$9.7 billion, serving over three million low-income New Yorkers annually. This means the **interpreter shortage Pinto described is a staffing-allocation decision inside a multi-billion-dollar budget**, not a lack of overall resources — useful framing for a pitch to the city (this is a *prioritization* problem, which is squarely a "what does the mayor care about" question per our advisor).
- **LA homelessness funding** is the clearest cautionary tale on the "who carries the risk" question. Los Angeles County is LAHSA's largest funding source, contributing roughly 46% of the agency's budget (about \$379 million), and the agency had an \$875 million budget in the most recent fiscal year before HUD suspended federal Continuum of Care funding pending an investigation. LAHSA was given 30 days from June 11, 2026 to request a formal hearing contesting HUD's findings and the suspension of Continuum of Care funding. The risk of that mismanagement fell first on the **frontline outreach workers and their clients** (motel-based transitional housing residents), not on the administrators who signed the contracts — echoing Erik's point that "the system doesn't actually help — it's the people doing the work."
- **MBPO / Community Boards**: funded as an independent city agency; Community Board members are **unpaid volunteers** (up to 50 per board, appointed by the Borough President), meaning the "labor" of local governance itself is donated. Community boards are composed of 50 volunteer members serving staggered two-year terms, and the Manhattan Borough President's Office appoints roughly 300 members to the borough's 12 community boards each year while providing professional support and training to increase their efficiency and impact.
- **Nonprofit/makerspace funding (Curren Mandon interview)**: grant-funded on a yearly/quarterly cycle requiring renewed grant proposals to keep operating; free certification programs are still staffed by paid instructors, creating constant budget "scramble." Funding partners also request demographic/outcome data from members, which nonprofits have to produce as a condition of continued funding — a second, quieter data-collection burden layered on top of service delivery. (*Interview 9/9*)
- **Roosevelt Island / Cornell Tech civic engagement**: appears to run on **almost no dedicated budget or staff** — Jane is described as the sole person "upkeeping, communicating" the relationship between Cornell Tech and the Island's institutions

(RIOC, Senior Center, Art Gallery, Hospital, PSI School), which is a resourcing gap as much as a process gap. (*Interview 9/10*)

Pain Points

Who is frustrated? What is inefficient, costly, or still nobody's job to fix?

Fragmented, incompatible systems

- Mr. Vargas (Homeless Services Ombudsman) named this directly: agencies have the services people need, they just "don't do it" well, largely because systems don't update or talk to each other (*Interview 9/11*)

Referral-only, non-proactive systems

- Both LA outreach (Erik) and, implicitly, NYC's 311 model rely on someone actively reporting a problem before anything happens.
- Erik explicitly wished for something less top-down and more like a live map of who's already out there and what's needed where ("Blue Sky" vision).
- Academic research backs this up as a structural, not anecdotal, problem: 311-style systems suffer from reporting bias, where some subpopulations report at systematically lower rates than others, and a NYU Marron Institute study of NYC housing-condition data found that despite greater objective and subjective need, low-income and minority neighborhoods are less likely to report street-condition or "nuisance" issues, while prioritizing more serious problems, which means self-reported city data embeds inequities that agencies then unintentionally act on as if they were neutral demand signals.
- Independent research in Boston reached the same conclusion via a totally different method — comparing resident reports to computer-vision detections. An analysis of roughly 187,000 citizen 311 reports against nearly 5,000 independent computer-vision defect detections in one Boston neighborhood found only 7.9% overlap between what residents reported and what algorithms detected, showing that resident-generated and sensor-generated data capture fundamentally different — and both incomplete — pictures of need. Earlier scholarship anticipated exactly this dynamic: researchers hypothesized that historically disadvantaged groups would be less likely to use 311-style crowdsourced systems and therefore less likely to benefit from the resource allocation those systems drive, a gap they expected to widen as reporting shifted from phone calls to smartphone apps. **This is arguably the single most important pain point for our HMW question** — the city's main tool for "learning what residents need" structurally under-hears the residents who may need the most.

Motivational problems

- Mr. K's observation — people "don't want to learn the technology," they want someone to do it for them and get frustrated when that's not offered.

Language access

- Pinto flagged a shortage of interpreters at DSS as the single biggest unmet need — explicitly *not* a technology problem, which is an important caution against over-indexing our solution on data/software fixes alone.

No feedback loop / invisible wins

- Mr. Vargas noted leadership doesn't hear about "wins," only escalations, distorting how agencies understand their own effectiveness.
- Robbie Dornbush's engaged-citizen interview echoed this from the resident side: follow-through is the hardest part of using 311 or contacting an office — no read receipts, no visibility into what's actually being done
- Generally, this erodes morale and willingness to report future issues ("hoping it will be solved").

Interagency coordination and ownership ambiguity

- Named explicitly by Marielle Ali/Andrew Chang as MBPO's single biggest pain point: when a constituent copies every relevant city agency on an email, **it becomes unclear who owns the response**, and identifying "who needs to do what, when, and how" consumes disproportionate staff time. This is structurally similar to Mr. Vargas's systems-incompatibility complaint — both are coordination failures between agencies, not failures of any single agency's service delivery.

Institutional trust deficits

- Erik described residents (particularly unhoused clients) as having "extreme trust issues," worsened concretely by the LAHSA fraud findings. This is now a live, unfolding case (HUD's suspension, a September 15, 2026 Congressional hearing) rather than a background concern, and shows how quickly a data/accountability failure becomes a *trust* failure that then further depresses resident reporting — compounding the underreporting problem above.

Power concentrated away from the people closest to the problem

- Erik's strongest "Blue Sky" point: administrative structures sit further from the ground truth than outreach workers, but hold the decision-making power; he'd like to see the flow of authority (and information) reversed so boots-on-the-ground observations drive resourcing rather than top-down directives.

Under-resourced, single-point-of-failure local liaisons

- Jane's situation on Roosevelt Island is a small-scale but sharp illustration: one person, no staff, an unstructured Google contact list, no relationship-tracking tool, and a looming December 2026 retirement with no formal handover process. She explicitly said she'd value a tool for **tracking relationships and gathering feedback after events** — a concrete, buildable pain point.

Awareness problems

- (9/4 MBPO debrief) flagged that our HMW question implicitly assumes residents *can* advocate for themselves, know what 311 is, and won't be discouraged by unresponsive government — assumptions the interviews above repeatedly complicate.

Appendix

Advisor Guidance

- **Advisor-recommended process for our own team (David Opatrny, 9/3):** watch the city (not residents) as it tries to get things done and capture those signals; frame around what the mayor's office already cares about (housing, transit, ridership, records); avoid a "build trap" — start with public APIs and lightweight Python ETL before anything more ambitious; get familiar with event-driven agent harnesses, "progressive crystallization" (turning ad hoc human judgment calls into deterministic code over time), and API auth for messy data. (*Interview 9/3*)
- **Coach-recommended process (Burke Reimann, 9/1, 9/8):** validate before building — do 4Ps research and 10+ logged interviews before committing to a build; pair high-level conversations with rapid mockups shown back to interviewees within days.

Open Questions

- How a request is routed internally or how data flows between 311 and destination agencies?
- Budget/funding details for **NYC 311 / OTI** itself. Who funds data infrastructure upgrades versus call-center staffing?
- **Overuse / power-user distortion.** What share of 311 complaints comes from a small number of repeat "power users," and does that further skew city agencies' sense of citywide need? This is a good, sharp question to bring to Joseph Morrisroe.
- Financial picture of the **MTA's data & analytics function**. Is this capital-budget funded, or absorbed into general operating costs, and whether "who carries the risk" of bad transit data (delays, safety) maps at all to the risk model we're seeing in social services.
- We don't yet know NYC DHS's specific budget breakdown (only HRA's) or how much of it is earmarked for the kind of system-interoperability fix Mr. Vargas described needing.
- **311NYC internal routing/process detail.** Our own interview guide from 9/10 was never actually filled in with answers — we're currently relying on secondary/historical sources (NYC.gov, news coverage, a 2003-era Nextgov piece) rather than a first-hand account of how 311 *currently* routes and prioritizes requests. **Joseph Morrisroe's interview should close this gap directly** — he has led 311 since 2006 and can speak to how routing, data quality, and inter-agency handoffs actually work today, not two decades ago.
- **311's current technology stack and budget.** We don't know what NYC currently spends on 311's backend, whether OTI is investing in AI-assisted triage, or how (if at all) 311 data feeds back into resource-allocation decisions at destination agencies (DSS, DHS, Parks, etc.) — good questions for Morrisroe.
- **Transit as a "north star" domain.** David Opatrny's advice was to align our north star with issues like housing, transit, and ridership — we currently have zero first-hand insight into transit. **Gayan Seneviratna's interview** should tell us how the MTA collects,

structures, and uses rider/service data; whether there's an equivalent of the 311 "reporting bias" problem in transit complaints (e.g., delay reports, safety concerns); and whether MTA data could plausibly correlate with the social-services friction points we've already found (e.g., does poor transit access correlate with DSS/DHS service gaps?).

- **NYC DHS-specific budget.** We have granular HRA budget data but not a DHS-specific breakdown; useful for sizing any homelessness-services intervention.
- **Quantified underreporting for NYC specifically.** The equity/bias research we found (Kontokosta et al., Marron Institute) focuses on NYC heating/hot-water complaints — we should look for whether comparable underreporting studies exist for other complaint categories (e.g., homelessness encampments, sanitation) that are closer to our likely build target.
- **How HMIS Clarity/Champ compare to NYC's own homeless-services data systems** — Erik's insight was LA-specific; we don't yet know whether NYC DHS uses a comparable (or shared, since HMIS is a federal HUD requirement) system, and whether NYC's version has the same referral-only, non-proactive limitations.
- **Validation of the "power user" hypothesis** for 311 — worth a direct question to Morrisroe and/or a quick pull of NYC Open Data's 311 Service Requests dataset, which is publicly available and updated daily.
- **MBPO's current relationship-tracking tools (or lack thereof).** Marielle Ali offered to share a write-up on the office's stakeholder relationships — we should follow up and request it, since it may reveal the same "no tooling" gap Jane described for Roosevelt Island, suggesting a pattern (not a one-off) worth designing around.

Sources

Interviews (Team 405 notes):

- DSS/Pinto, Mr. Vargas, Mr. K, Career & Training Services security guard (9/11); Advisor meeting (9/11); Erik, non-profit outreach LA (9/10); Jane, RIOCC/Cornell Tech (9/10); 311NYC (9/10, unanswered); Curren Mandon, Makerspace (9/9); Robbie Dornbush, engaged citizen (9/10); Coach Burke Reimann (9/1, 9/8); Andrew Chang & Marielle Ali, MBPO (9/4); Advisor David Opatrny (9/3).

Secondary research:

- NYC.gov 311 "Best Practice: Call Center for Non-Emergency City Services"; NYC Council Data Team, "311 Agency"; citymeetings.nyc, NYC Council Committee on Governmental Operations (3/1/2024); 99% Invisible, "What Happens When I Call 311?"; White & Trump, "The Promises and Pitfalls of 311 Data," *Urban Affairs Review* (2018); Boxer, Hong, Kontokosta & Neill, "Estimating Reporting Bias in 311 Complaint Data," *Annals of Applied Statistics* (2025); Kontokosta & Hong, "Bias in Smart City Governance," NYU Marron Institute; O'Brien, "How Public Data Can Improve Services and Empower Communities," Scholars Strategy Network; Clark, Jang & Brudney, "Socio-Economic Status and Citizen Participation in Crowdsourced Government" (2011);

Harvard Kennedy School Data-Smart City Solutions, "When Residents and Algorithms See Different Problems"; NYC DSS/HRA Preliminary Budget Testimonies (FY18–FY26); HUD press release and news coverage on the LAHSA funding suspension (June–September 2026); NYC Charter §2800 / §82(17)(a); Manhattan Borough President's Office community board reports and press releases (2026).